

Critique of ICM — Praxis Sprint UE1

Part 2 — Critique of ICM

What critique did you find? Why is it important?

ICM's strong TruthfulQA result is partly explained by memorization, not purely by mutual predictability over coherent labelings. TruthfulQA has been in many model training corpora since its 2021 release; rewarding "mutual predictability" on a contaminated dataset partially rewards models that have seen the questions and their canonical answers during training. The reported result conflates "the model has coherent beliefs about truth" with "the model has memorized which answers are labeled true."

I also considered two other angles. **Coherence $\neq$ truth** — ICM could produce coherent-but-systematically-wrong labels if the base model's training data is biased; I set this aside because the experiment needs a domain with known model bias, which is harder to construct in a sprint. **Single-batch arbitrariness** — ICM defines coherence within a batch but runs independently across batches, so different chunking could change labels; I set this aside because the implication for the paper's headline claims is weaker than the memorization concern.

This critique matters because ICM's claimed safety applications — eliciting latent knowledge, honesty training, value specifications — all require the method to surface beliefs the model has but humans haven't labeled. If memorization explains a meaningful share of the TruthfulQA result, those applications don't work as advertised: the method would be retrieving labels the model has already seen, not eliciting latent beliefs.

Test design and preliminary results

Note on platform constraint. Within the sprint timeframe I ran preliminary versions of Experiments 1 and 2 below using a *proxy* for ICM (zero-shot direct classification with a chat model) rather than the full ICM search. The reason is that Hyperbolic blocked top-K logprob access on Llama-3.1-405B, so the full ICM simulated-annealing loop could not be run end-to-end. The zero-shot baseline is a meaningful proxy: it measures an *upper bound* on how much memorization could be contributing to the reported ICM result. If the model can already classify TruthfulQA correctly with no ICM at all, ICM has very little additional work to do, and the headline result is dominated by what the chat model already knows from pretraining.

Experiment 1 — paraphrase test on TruthfulQA. Take 30 TruthfulQA questions, run zero-shot direct classification with the chat model (Llama-3.3-70B-Instruct), paraphrase the questions while preserving semantic content, and re-run classification. Compare label agreement. If the model’s judgment is over the underlying truth-judgment, paraphrasing should change labels minimally. A large drop is a memorization signature — the model was anchoring on lexical features of the original questions, not their semantic content.

Experiment 2 — control for the paraphrasing-in-general effect. Experiment 1 alone is confounded: paraphrasing itself might destabilize classification regardless of memorization. So I ran the same paraphrase-and-re-label procedure on 30 questions we wrote ourselves, post-training-cutoff, on topics with verifiable answers. Compare the label-drop on TruthfulQA against the label-drop on this control set.

Observed results. Preliminary results from running this test (n=30 per condition) showed exactly the pattern memorization predicts. TruthfulQA accuracy dropped from 96.7% to 83.3% (−13.3 pp) under paraphrasing. The post-cutoff control dropped only 63.3% → 60.0% (−3.3 pp). The ~10 pp gap is the memorization-specific contribution. That share of the original ICM result is not attributable to coherence-over-truth — it’s attributable to memorization of the original surface form of TruthfulQA’s questions.

With more time. Repeat on multiple model families (LLaMA, Mistral, Gemma) and scales (1B, 7B, 70B) to test whether memorization explains more at larger scale, where memorization capacity is higher. Extend the post-cutoff control set to 200+ questions for tighter confidence intervals. Add a third arm where a *different* model family generates the paraphrases, to break the circularity discussed below.

Honest limitations. The same model both paraphrases the questions and judges them, which is circular: a model that has memorized a question may paraphrase it in a way that re-triggers its own memorized response, hiding the memorization signal. A cleaner protocol would use a different model family for paraphrase generation. Additionally, our control set is human-written, which risks a difficulty mismatch with TruthfulQA: if our questions are easier on average, the paraphrasing-induced drop will be smaller for reasons unrelated to memorization. Mitigation: pilot the control set on the base model to confirm baseline accuracy is comparable, and match TruthfulQA’s question style and topical distribution as closely as possible.